What is student learning?

What happens inside their heads when they explore a topic, as they think about what the words mean, and try and assimilate some knowledge for future use.

Our objectives?

- Help students acquire knowledge through their learning;
- Help students develop their ability to learn new things;
- Help students acquire specific competencies/skills;
- Help students develop confidence?

Most of these are very adventurous in a lecture format!

What are we trying to engage students *with*?

- We try to engage students with the lectures / classes, tutorials, provided homeworks, and other activities;
- We try to engage students with their own learning.

Bad news

Just making students turn up, listen, pay attention, and even do exercises and perform tasks might not be sufficient, but it's probably our goal most of the time.

Good news

We don't have to do all the work. You can't put learning in someone's head. Co-creating a Learning Community helps students own their own learning.

An important principle

Decide the objectives of activities we design: lectures, homeworks, courseworks and communicate these objectives to the students.

Active learning techniques – in lectures

Low to moderate overhead

- Coloured voting cards;
- Annotations (marginal note-taking during lectures);
- (Tech-based) Interactive polling: Socrative, Pingo, Mentimeter, Turning Point.

Active learning techniques – in lectures

Moderate to high overhead

- Gapped notes, i.e. purposeful gaps;
- (Tech-based) Demonstrations / Interactive software: Geogebra, Wolfram, RShiny, etc.;
- In-class group discussions: prepared activities / questions.

Active learning techniques

Substantial overhead

- Peer-assessment (with student training);
- Full class redesign, e.g. inversion (see later).

What is Question-Driven Instruction?

- Question-Driven Instruction is Peer Learning with the use of in-class polling.
- Technology just a convenient tool for large classes.
- The key educational benefit (as indicated in rich literature since ≈2000) is focussed on the benefits of lecturer mediated Peer Learning of difficult concepts.
- Question design is a key element.

Student comments

Lectures are much more engaging than other modules, it makes the content way more interesting and makes me actually not mind attending a 2 hour 9am. :)

I could only wish to have this style of lecture sooner. Every lecture is rewarding and answers all my questions I have whilst reading my notes beforehand. If every lecture was like this it would make the 9000 pounds worth it.

I submitted a feedback form within the first couple weeks of the lectures and I was very unsure about the style of the module. However as the weeks have gone on I have felt a little more engaged and have liked reading the notes before each week. My main concern is that I would like it if the questions in lectures could be linked to exam questions more easily.

Student comments

I really don't like this style of lectures. The lecture notes are good but usually I like the lecturer explaining on top of the lecture notes [...]. 'Normal' lectures allows for a better explanation of the material we will be assessed on.

I think that the idea of question driven learning is a good one, however, it is very difficult to get all the reading done on top of all the other modules so it would be nice if you could spend 5 minutes at the beginning of each lecture reminding us of the key points of key formulae which we should have learnt about from our reading [...].

I appreciate the new technique of teaching however feel as though it would be helpful if you could refer to the lecture notes more. If possible could you read and follow a set of lecture notes to us like in a 'normal lecture', then have us answer questions for the last 25 mins. This way we can still learn and be tested at the same time